

1. Given two sets $A = 1, 2$, $B = 1, 2, 3$, then ().
 (A) $A = B$ (B) $A \subseteq B$ (C) $A \in B$ (D) $B \subseteq A$
2. Given two sets $A = (-1, 3)$, $B = [2, 4]$, then $A \cap B = ()$.
 (A) $(2, 3]$ (B) $[2, 3]$ (C) $[2, 3)$ (D) $(2, 3)$
3. The solution set of the inequality $x^2 - 5x + 4 < 0$ is ().
 (A) $(-\infty, 1) \cup (4, +\infty)$ (B) $(-\infty, -4) \cup (-1, +\infty)$
 (C) $(-4, -1)$ (D) $(1, 4)$
4. The domain of the function $y = \frac{1}{\sqrt{x+1}}$ is ().
 (A) $[-1, +\infty)$ (B) $(-1, +\infty)$ (C) $(0, +\infty)$ (D) $(-\infty, -1)$
5. Let a_n be an arithmetic sequence, with the first term $a_1 = 2$ and common difference $d = 3$. Then $a_{100} = ()$.
 (A) 305 (B) 299 (C) 298 (D) 302
6. In the planar rectangular coordinate system, the point $(-1, 3 - \pi)$ belongs to ().
 (A) the second quadrant (B) the third quadrant
 (C) the fourth quadrant (D) the first quadrant
7. $\left(\sin \frac{\pi}{4} + \cos \frac{\pi}{4}\right) \times \tan \frac{\pi}{4} = ()$.
 (A) 1 (B) $2\sqrt{2}$ (C) 2 (D) $\sqrt{2}$
8. Which of the following functions is odd? ()
 (A) $y = x^4$ (B) $y = x \sin x$ (C) $y = \cos x$ (D) $y = x + x^3$
9. The inverse function of the function $y = x^3$ is ().
 (A) $y = \sqrt[3]{x}$ (B) $y = \frac{x}{3}$ (C) $y = x^3$ (D) $y = 3^x$
10. Which of the following functions is monotonic increasing? ()

(A) $y = \left(\frac{1}{3}\right)^x$ (B) $y = \log_{0.5} x$ (C) $y = 2x + 1$ (D) $y = \sin x$

11. Which of the following points is in the third quadrant? ()

(A) (1,2) (B) (-1,2) (C) (-1,-2) (D) (1,-2)

12. The solution set of the rational inequality $\frac{x-1}{x-2} \leq 1$ is ().

(A) [1,2] (B) $(-\infty, 1)$ (C) $(-\infty, 2)$ (D) $(2, +\infty)$

13. Given that a , b and c form an arithmetic sequence that satisfies $a + c = 20$, then $b = ()$.

(A) 10 (B) 20 (C) 0 (D) 5

14. The angle of inclination of the line $y = \sqrt{3}x + 10$ is ().

(A) 120° (B) 60° (C) 30° (D) 45°

15. Which of the following inequalities is correct? ()

(A) $0.75^{-0.2} > 0.75^{-0.4}$ (B) $3^{2.25} > 3^3$ (C) $2.1^{\frac{2}{3}} > 1.2^{\frac{2}{3}}$ (D) $2.1^{-2} > 1.2^{-2}$

16. Suppose that the point $P(1, y)$ is on the line that makes an angle α with the x -axis, and $\tan \alpha = 2$. Then $y = ()$.

(A) 1 (B) 2 (C) $\frac{1}{2}$ (D) $\sqrt{3}$

17. Given that a_n is an arithmetic sequence, with $a_1 = 1$, $a_2 = 2$, then $a_{2025} = ()$.

(A) 20
(B) 2024
(C) 2
(D) 2025

18. A circle has center at $(-3, 2)$ and radius 4. Then the equation of the circle is ().

(A) $(x + 3)^2 + (y - 2)^2 = 4$
(B) $(x - 3)^2 + (y + 2)^2 = 4$
(C) $(x - 3)^2 + (y + 2)^2 = 16$

(D) $(x+3)^2 + (y-2)^2 = 16$

19. The distance from a point $P(-1,-1)$ to another point $Q(1,1)$ is ().

- (A) 4 (B) $2\sqrt{2}$ (C) 2 (D) $\sqrt{2}$

20. The equation of the line passing through the points $P(1,2)$ and $Q(2,4)$ is ().

- (A) $y = x + 2$ (B) $y = 2x$ (C) $y = x + 1$ (D) $y = 2x + 2$

21. Given the equation of a circle $x^2 + y^2 + 2x - 2y = 0$, the radius of the circle is ().

- (A) 2 (B) $\sqrt{2}$ (C) $\frac{1}{2}$ (D) 1

22. If α is an acute angle and $\sin\alpha = \frac{\sqrt{3}}{2}$, then $\tan\alpha = ()$.

- (A) $\sqrt{3}$ (B) $\frac{1}{4}$ (C) $\frac{\sqrt{3}}{3}$ (D) $\frac{1}{2}$

23. Given that the first three terms of a geometric sequence $\{a_n\}$ are 1, 2, and 4, the general formula of the sequence is ().

- (A) $a_n = 8$ (B) $a_n = 2^n$ (C) $a_n = 2n$ (D) $a_n = 2^{n-1}$

24. Given that $a > b$, then ().

- (A) $5^a > 5^b$ (B) $a^2 > b^2$ (C) $\sin a > \sin b$ (D) $\frac{1}{a} < \frac{1}{b}$

25. The distance between the points $P(-1,2)$ and $Q(3,1)$ is ().

- (A) $\sqrt{17}$ (B) $\sqrt{5}$ (C) $\sqrt{13}$ (D) 5

26. Which of the following descriptions about the function $y = \left(\frac{1}{2}\right)^x$ is correct? ()

- (A) The function is monotonic decreasing
(B) The function is symmetric about the origin
(C) The function is symmetric about the x axis
(D) The function is monotonic increasing

27. If $\sin\alpha = \frac{1}{4}$, then $\cos 2\alpha = ()$.

- (A) $-\frac{7}{8}$ (B) $\frac{7}{8}$ (C) $-\frac{1}{8}$ (D) $\frac{1}{8}$

28. The intersection of the two lines $l_1: y = 2x + 1$ and $l_2: x + y + 1 = 0$ is ().

- (A) $\left(-\frac{1}{3}, -\frac{2}{3}\right)$ (B) $\left(\frac{2}{3}, \frac{1}{3}\right)$ (C) $\left(-\frac{2}{3}, -\frac{1}{3}\right)$ (D) $\left(\frac{1}{3}, \frac{2}{3}\right)$

29. The coordinates of the foci of the hyperbola $x^2 - y^2 = 1$ are ().

- (A) $(0, \pm 2)$ (B) $(\pm \sqrt{2}, 0)$ (C) $(\pm 2, 0)$ (D) $(0, \pm \sqrt{2})$

30. $\log_2 32 + \log_{\frac{1}{2}} 32 = ()$.

- (A) -5 (B) 10 (C) 0 (D) 5

31. Which of the following descriptions about the function $y = \cos x$ is correct? ()

- (A) The function is periodic with period 2π
(B) The function is monotonic increasing over the interval $[0, \pi]$
(C) The function is odd
(D) $\cos(x + \pi) = \cos x$

32. About the parabola $y^2 = 4x$, which of the following statements is correct? ()

- (A) The coordinates of the foci are $(\pm 1, 0)$
(B) The axis of symmetry is $x = 0$
(C) The coordinates of the focus are $(1, 0)$
(D) Passes through the point $(-4, 4)$

33. The line that is perpendicular to $l: x + y + 3 = 0$ is ().

- (A) $y = -2x + 1$ (B) $y = 2x + 1$ (C) $y = x + 3$ (D) $y = 1 - x$

34. Suppose that $\cos \alpha = -\frac{1}{2}$, $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, then the value of $\sin \frac{\alpha}{2}$ is ().

- (A) $\frac{\sqrt{3}}{2}$ (B) $-\frac{\sqrt{3}}{2}$ (C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

35. Given that $\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$ and $\sin \frac{\pi}{6} = \frac{1}{2}$, using the formula $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$, one can obtain that $\cos \frac{\pi}{12} = ()$.

(A) $\frac{\sqrt{6}-\sqrt{2}}{4}$ (B) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (C) $\frac{1+\sqrt{3}}{4}$ (D) $\frac{\sqrt{6}}{4}$

36. The function that is equivalent to $y = |x|$ is ().

(A) $y = \sqrt{x^2}$ (B) $y = x$ (C) $y = (\sqrt{x})^2$ (D) $y = \sqrt[3]{x^3}$

37. If α is an acute angle, with $\cos\alpha = \frac{1}{2}$, then $\sin\frac{\alpha}{2} = ()$.

(A) $\frac{1}{2}$ (B) $\pm\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\pm\frac{\sqrt{3}}{2}$

38. Which one of the following reduction formulae is false? ()

(A) $\sin\left(\frac{\pi}{2} - \alpha\right) = \cos\alpha$

(B) $\sin(\pi - \alpha) = \sin\alpha$

(C) $\tan(\pi + \alpha) = -\tan\alpha$

(D) $\cos(2\pi - \alpha) = \cos\alpha$

39. The equation of the directrix of the parabola $y^2 = -x$ is ().

(A) $x = \frac{1}{4}$ (B) $x = \frac{1}{2}$ (C) $x = -\frac{1}{2}$ (D) $x = -\frac{1}{4}$

40. Given that $\frac{\cos\alpha - \sin\alpha}{\cos\alpha + \sin\alpha} = \frac{1}{3}$, then $\tan\alpha = ()$.

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

41. If a sequence a_n satisfies $a_1 = 1$, $a_n = \frac{1}{1 + \frac{1}{a_{n-1}}}$ ($n \geq 2$), then $a_{100} = ()$.

(A) 50 (B) 100 (C) $\frac{1}{100}$ (D) 1

42. About the logarithmic function $y = \log_a x$, which of the following statements is correct? ()

(A) When $a > 1$, $\log_a x > 1$ for any $x > 0$

(B) The $a > 1$, the function is monotonic decreasing over the domain

(C) The domain of the function is $(-\infty, +\infty)$

(D) When $0 < a < 1$, the function is monotonic decreasing over the domain

43. About the ellipse $\frac{x^2}{4} + \frac{y^2}{20} = 1$, which of the following statements is correct? ()

- (A) The two foci are $(0, \pm 4)$
- (B) The vertices are $(\pm 4, 0)$ and $(0, \pm 20)$
- (C) The major and minor axes have lengths 20 and 4 respectively
- (D) The two foci are $(\pm 4, 0)$

44. Suppose that a straight line l is perpendicular to another line $l_1: x + 2y + 4 = 0$ and passes through the intersection of $l_2: x + y + 1 = 0$ and $l_3: 2x + y - 1 = 0$. The equation of l is ().

- (A) $x + 2y - 3 = 0$ (B) $x - 2y - 1 = 0$ (C) $2x + y - 5 = 0$ (D) $2x - y - 7 = 0$

45. Let O be the origin. A and B are two points on the plane, and P is the midpoint of AB . Then the vector $\overrightarrow{OP}$ can be represented by ().

- (A) $\frac{1}{2}(\overrightarrow{OA} + \overrightarrow{OB})$ (B) $\overrightarrow{OA} + \overrightarrow{OB}$ (C) $\overrightarrow{OB} - \overrightarrow{OA}$ (D) $\frac{1}{2}(\overrightarrow{OB} - \overrightarrow{OA})$

46. If a complex number z satisfies $z^3 = 1$ and $z \neq 1$, then $1 + z + z^2 + \cdots + z^{999} = ()$.

- (A) $\frac{1+\sqrt{3}i}{2}$ (B) $\frac{-1+\sqrt{3}i}{2}$ (C) i (D) 1

47. Given that an arithmetic sequence $\{a_n\}$ satisfies $3(a_2 + a_6) + 2(a_6 + a_{10} + a_{14}) = 24$, then the sum of the first 13 terms of $\{a_n\}$ is $S_{13} = ()$.

- (A) 104 (B) 26 (C) 39 (D) 52

48. The Chinese zodiac, also known as the birth animal sign, consists of twelve animals representing different years in a repeating cycle. The order is: Rat, Ox, Tiger, Rabbit, Dragon, Snake, Horse, Goat, Monkey, Rooster, Dog, and Pig. For example, those born in 1984, 1996, or 2008 are Rats, while those born in 1985, 1997, or 2009 are Oxen, and so on. If three people are randomly selected, the probability that at least two of them share the same zodiac sign is ().

- (A) $\frac{1}{12}$ (B) $\frac{143}{144}$ (C) $\frac{55}{72}$ (D) $\frac{17}{72}$